Sahaj Bhadja

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Summary

Software engineer with experience building **data processing pipelines**, **internal tools**, and backend systems for large-scale analysis. Proficient in **Python, Go, and C++** with exposure to **AWS/GCP**, **PostgreSQL**, and analytics workflows. Passionate about **autonomous driving technology** and simulation-driven validation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Distributed Systems, Machine Learning, Analysis of Algorithms, Database Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Data Structures, Operating Systems, Robotics Fundamentals, Software Engineering

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Contributed to **data processing pipelines** and internal services supporting large-scale evaluation workflows, writing reliable code with **testing, code reviews, and CI**.
- Built backend tooling in **Python and Go** to process high-volume datasets, debug performance issues, and improve operational reliability for production systems.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **internal tools and dashboards** using **Node.js, React, and PostgreSQL** for test execution tracking, analysis, and reporting across distributed teams.
- Deployed services on **AWS/GCP** with Docker, partnering across the stack to ship maintainable features in a fast-paced engineering environment.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **simulation and analysis pipelines** for robotics/ML experiments, processing driving-style trajectory data and generating metrics for validation workflows.
- Improved pipeline throughput by 18% through performance tuning and scientific analysis of large-scale evaluation outputs stored in **PostgreSQL**.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Designed backend systems and **ETL pipelines** in **Python and C++** for real-time data ingestion, transformation, and reporting across distributed nodes.
- Supported operational excellence through monitoring, debugging, and version-controlled releases using Git and automated testing practices.

Mobility Systems Lab **India**
Robotics Software Intern *January 2023 – April 2023*
- Contributed to **AV/ADAS simulation** tooling in **Python and C++**, helping organize test runs, analyze scenario outcomes, and generate release-readiness metrics.
- Assisted with developer infrastructure for test triggering and result analysis, gaining hands-on exposure to data-driven engineering workflows.

Skills

Languages: Python, Go, C++, JavaScript, SQL, Bash
Systems: Data pipelines, distributed systems, backend services, testing, debugging, Git, CI/CD
Data & Cloud: PostgreSQL, BigQuery basics, AWS, GCP, Docker, analytics workflows
Web & Tools: Node.js, React, internal dashboards, developer tooling

Projects

DriveEval | *Python, Go, PostgreSQL, AWS*
- Built a **simulation evaluation platform** for analyzing large volumes of driving scenario data, supporting test organization, metric computation, and release reporting workflows.

SimMetrics | *Python, C++, GCP, BigQuery*
- Developed **data processing pipeline** for ADAS-style validation, ingesting test logs and producing analytics dashboards for stack verification and performance tracking.

Big-Brain Live | *Python, React, Node.js, PostgreSQL*
- Created internal analysis tool with React dashboard and Python backend for large-scale document evaluation, demonstrating end-to-end data workflow development.